



Configuring SNMP and the Log Service

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

Contents

1 About This Lab	2
1.1 Task Description	2
1.2 Objectives	2
2 Task Preparation	2
2.1 Network Topology	2
2.2 Initial Configuration	2
3 Task Implementation	3
4 Verification	10
5 Quiz	11



1 About This Lab

1.1 Task Description

Simple Network Management Protocol (SNMP) is a standard protocol widely used for managing network elements (NEs) on TCP/IP networks. It enables centralized management through a central computer running network management software, that is, a network management station (NMS). SNMP supports three versions: SNMPv1, SNMPv2c, and SNMPv3. One or more versions can be configured as needed.

Syslog is an industrial standard protocol used to record device logs. System logs record all events that occur in the system at any time on UNIX systems, routers, and switches. Administrators can use these logs to learn about the system status and diagnose faults.

After completing this task, you will have a clear understanding of SNMP and syslog and be able to use SNMP and syslog to collect device logs.

1.2 Objectives

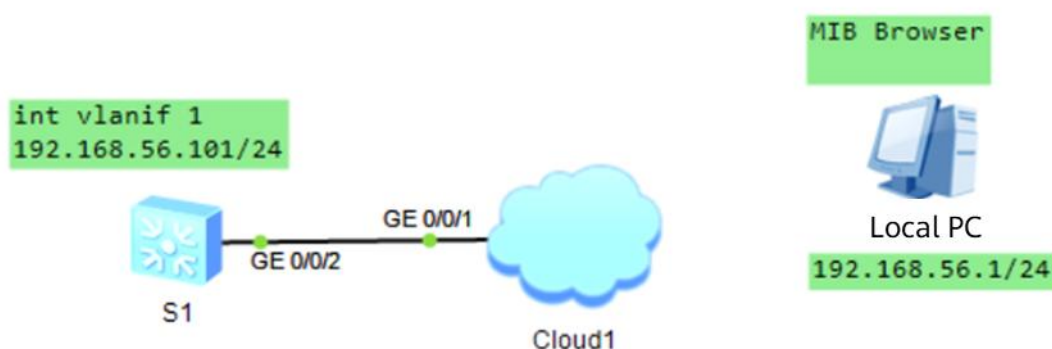
On completing this task, you will be able to:

1. Configure SNMP on a device.
2. Be familiar with the MIB Browser.
3. Configure syslog on a device.
4. Use tools to collect device logs.

2 Task Preparation

2.1 Network Topology

Figure 1 Configuring SNMP and the log service



2.2 Initial Configuration

- The initial configuration of S1 is as follows.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
```



```
[Huawei]sysname S1
[S1]int Vlanif 1
[S1-Vlanif1]ip address 192.168.56.101 24
[S1-Vlanif1]quit
[S1]
```

- The configuration of Cloud1 is shown in Figure 2.

Figure 2 Cloud1 configuration

The screenshot shows the 'Cloud1' configuration window with the 'IO 配置' (IO Configuration) tab selected. It includes sections for '端口创建' (Port Creation) and '端口映射设置' (Port Mapping Settings).

端口创建 (Port Creation):

- 绑定信息: VirtualBox Host-Only Network -- IP: 192.168.56.1
- 警告: 请勿绑定公网网卡, 否则可能会引起网络瘫痪。
- 端口类型: GE
- 监听端口: 30000
- 对端IP: 0.0.0.0
- 对端端口: 0
- 建议: (30000-35000)
- Buttons: 增加, 删除

No.	端口类型	端口编号	UDP端口号	端口开放状态	绑定信息
1	GE	1	61146	Internal	UDP
2	GE	2	None	Public	VirtualBox Host-Only Network -- IP: 192.168.56.1

端口映射设置 (Port Mapping Settings):

- 端口类型: GE
- 入端口编号: 1
- 出端口编号: 2
- ☒ 双向通道
- Buttons: 增加

端口映射表 (Port Mapping Table):

No.	入端口编号	出端口编号	端口类型
1	1	2	GE
2	2	1	GE

Buttons: 删除

3 Task Implementation

1. Checking Basic Configurations

Run a ping command on the local PC to check its connectivity with S1.

```
C:\Users\Administrator>ping 192.168.56.101
```

Pinging 192.168.56.101 with 32 bytes of data:

Reply from 192.168.56.101: bytes=32 time= 13ms TTL=255

Reply from 192.168.56.101: bytes=32 time= 4ms TTL=255

Reply from 192.168.56.101: bytes=32 time= 18ms TTL=255

Reply from 192.168.56.101: bytes=32 time= 6ms TTL=255

Ping statistics for 192.168.56.101:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss);

Approximate round trip times in milli-seconds:



Minimum = 4ms, Maximum = 18ms, Average = 10ms

C:\Users\Administrator>

2. Configuring SNMP on S1

- (1) Configure SNMPv2c on S1 so that the NMS using SNMPv2c can manage S1.

```
[S1]snmp-agent sys-info version v2c
```

```
[S1]
```

- (2) Configure an interface for receiving and responding to NMS requests.

```
[S1]snmp-agent sys-info version v2c
```

```
[S1]
```

- (3) Configure MIB views to specify the MIB objects that can be accessed by the NMS.

Configure the **system** subtree to be accessible in the MIB view **isoview01**.

```
[S1]snmp-agent mib-view included isoview01 system
```

Configure the **interfaces** subtree to be accessible in the MIB view **isoview01**.

```
[S1]snmp-agent mib-view included isoview02 interfaces
```

Configure the **ip** subtree to be accessible in the MIB view **isoview01**.

```
[S1]snmp-agent mib-view included isoview03 ip
```

- (4) Configure a community string. When a switch is added to the NMS, the community string is used for authentication and the ACL is applied for access control.

```
[S1]
```

Configure adminnms01 to have the read-only permission on the **system** subtree.

```
[S1]snmp-agent community read adminnms01 mib-view isoview01
```

Configure adminnms02 to have the read-only permission on the **interfaces** subtree.

```
[S1]snmp-agent community write adminnms02 mib-view isoview02
```

Configure adminnms03 to have the read-only permission on the **ip** subtree.

```
[S1]snmp-agent community read adminnms03 mib-view isoview03
```

```
[S1]
```

- (5) Configure a trap host and enable S1 to proactively send traps.

```
[S1]snmp-agent trap enable
```

Warning: All switches of SNMP trap/notification will be open. Continue? [Y/N]:y

```
[S1]
```

3. Reading MIBs on S1 Using the MIB Browser

- (1) Configure the MIB Browser and read the **system** subtree of S1, as shown in Figure 3 and Figure 4.



Figure 3 Configuring the MIB Browser

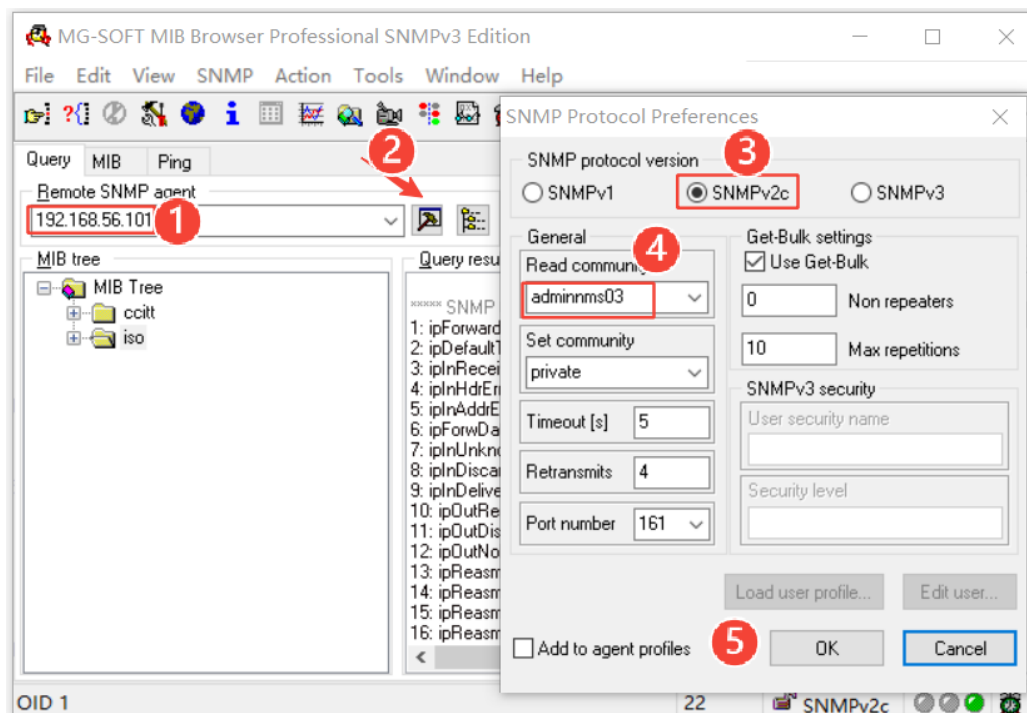
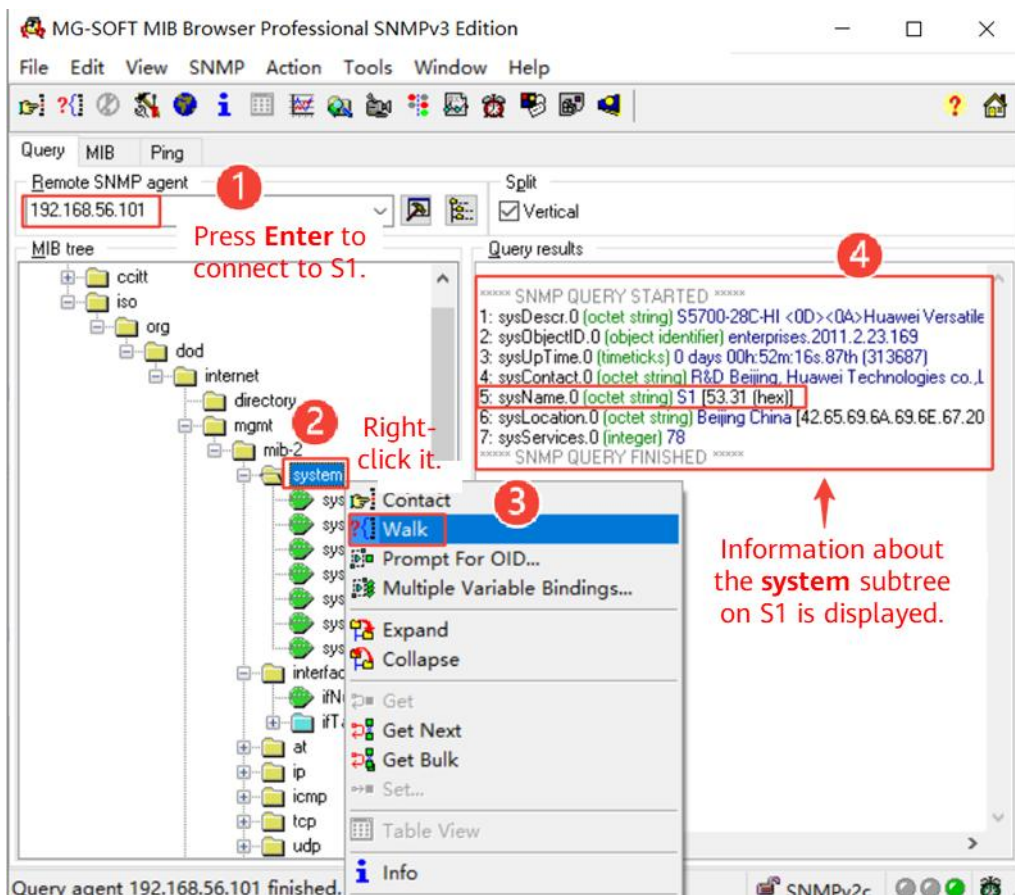


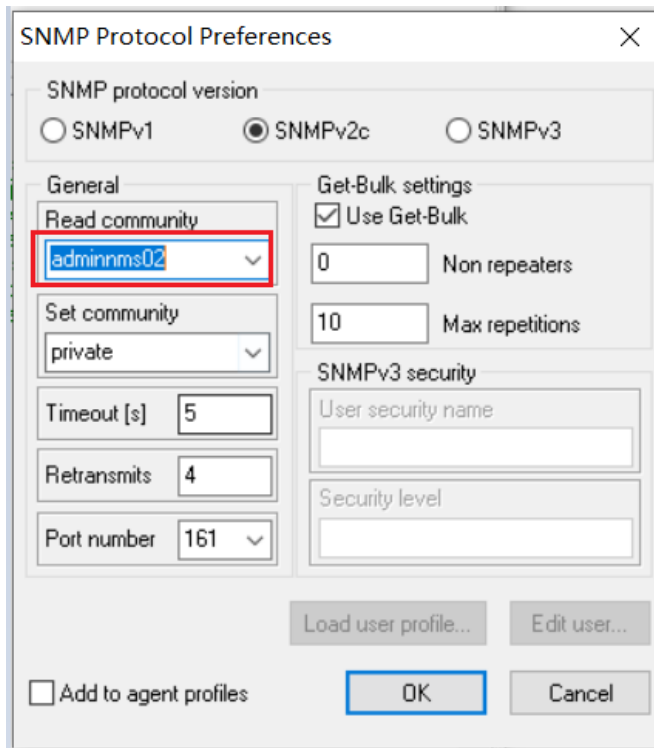
Figure 4 Reading the system subtree of S1





- (2) Configure the MIB Browser and read the **interfaces** subtree of S1, as shown in Figure 5 and Figure 6.

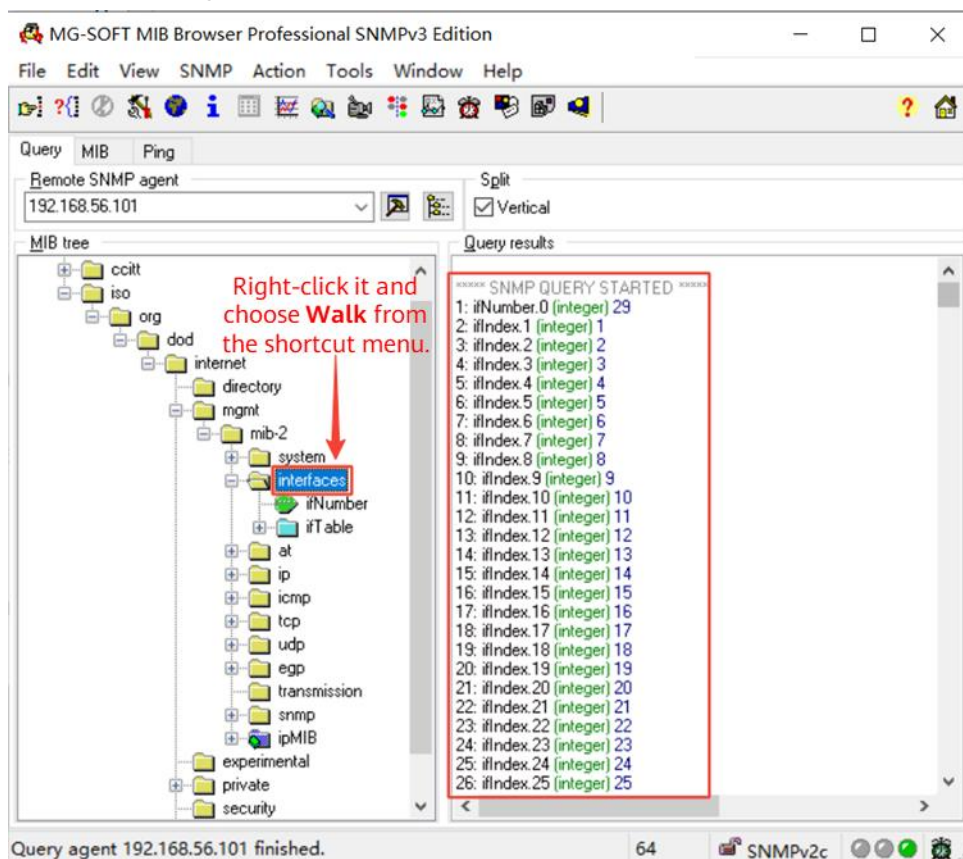
Figure 5 Configuring the read community of the interfaces subtree



The image shows the 'SNMP Protocol Preferences' dialog box. The 'SNMP protocol version' section has three radio buttons: 'SNMPv1', 'SNMPv2c' (which is selected), and 'SNMPv3'. The 'General' section contains a 'Read community' dropdown menu with 'adminnms02' selected, a 'Set community' dropdown menu with 'private' selected, a 'Timeout [s]' text box with '5', a 'Retransmits' text box with '4', and a 'Port number' dropdown menu with '161'. The 'Get-Bulk settings' section has a checked 'Use Get-Bulk' checkbox, a 'Non repeaters' text box with '0', and a 'Max repetitions' text box with '10'. The 'SNMPv3 security' section has a 'User security name' text box and a 'Security level' text box. At the bottom, there are buttons for 'Load user profile...', 'Edit user...', 'OK', and 'Cancel'. A checkbox labeled 'Add to agent profiles' is also present.

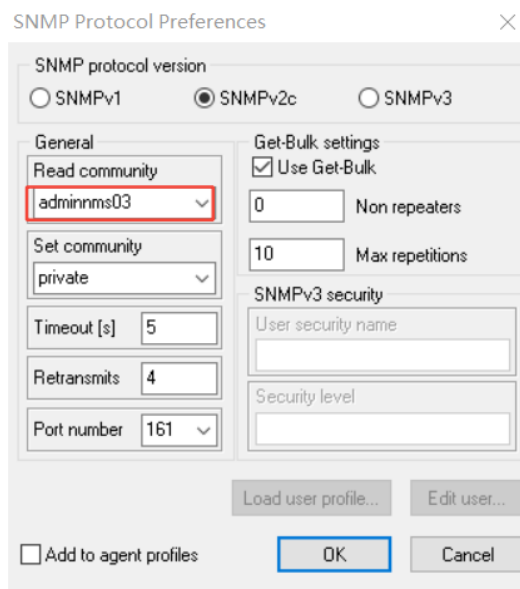


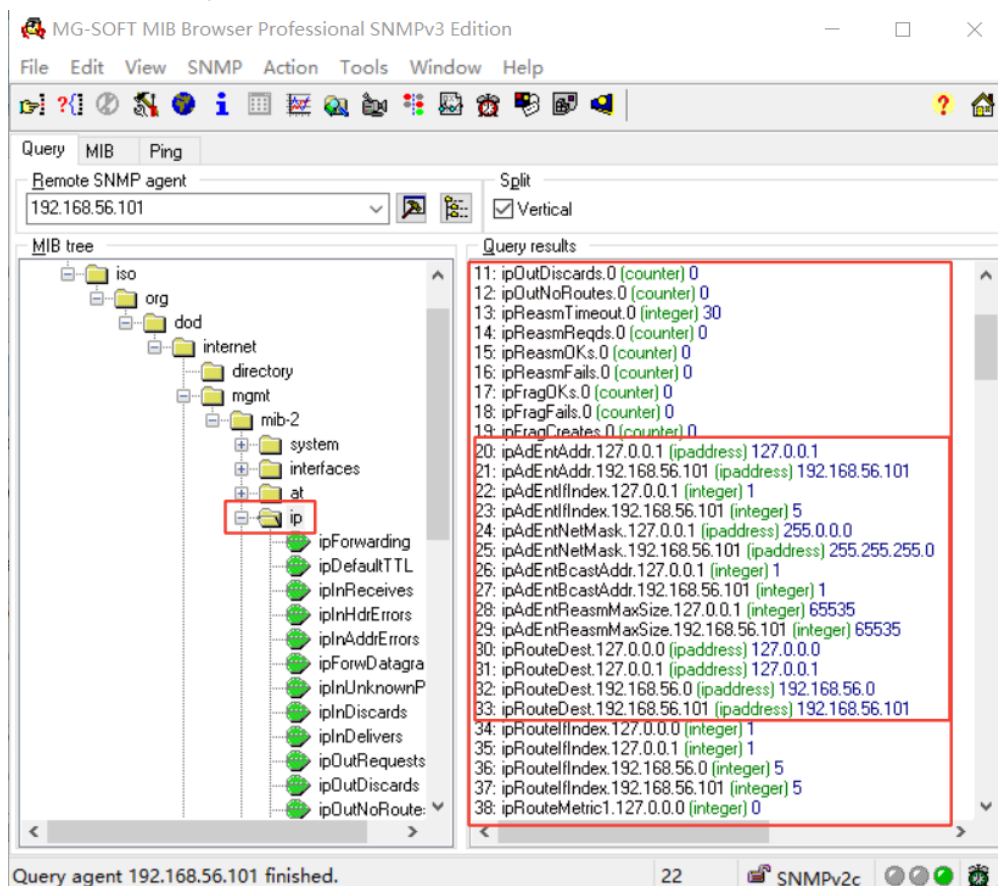
Figure 6 Reading the interfaces subtree



- (3) Configure the MIB Browser and read the **ip** subtree of S1, as shown in Figure 7 and Figure 8.

Figure 7 Configuring the read community of the ip subtree



**Figure 8** Reading the ip subtree

4. Configuring the Syslog Service on S1

Enable the information center function and configure SW1 to send notification logs generated by the ARP module to Server1.

```
[SW1]info-center enable
```

```
Info: Information center is enabled.
```

```
[SW1]info-center channel 6 name loghost1
```

```
[SW1]info-center loghost 192.168.56.1 channel loghost1
```

```
[SW1]info-center source arp channel loghost1 log level notification
```

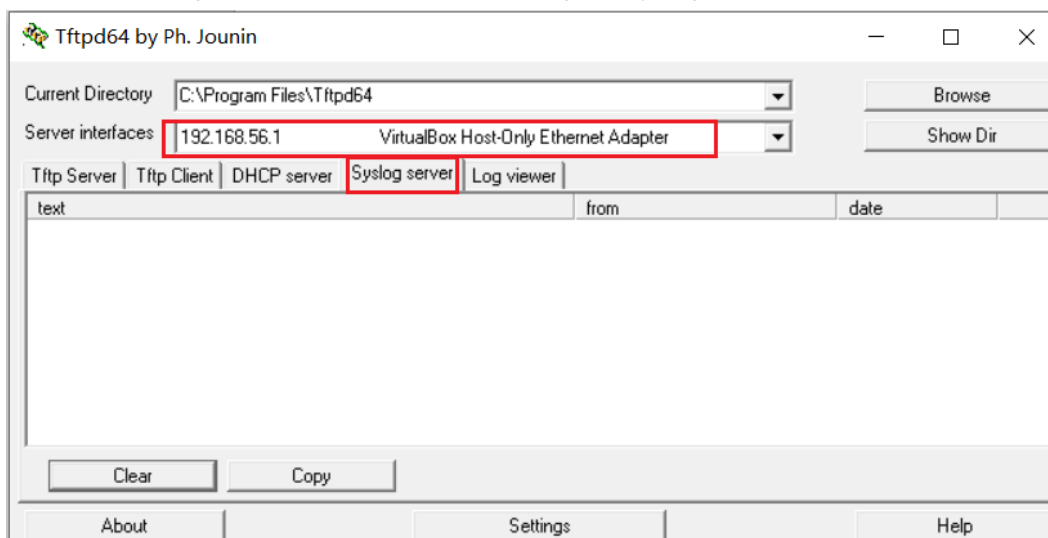
```
[SW1]
```

5. Checking Syslog Information Using Tftpd64 Software

(1) Open Tftpd64 and enable the syslog server, as shown in Figure 9. **Ensure that the host firewall is disabled.**



Figure 9 Starting Tftpd64 software and enabling the syslog server



- (2) On SW1, ping an IP address that does not exist. SW1 then initiates an ARP request.

```
[SW1]ping 10.1.1.3
  PING 10.1.1.3: 56 data bytes, press CTRL_C to break
    Request time out
    Request time out
    Request time out
    Request time out
    Request time out

  --- 10.1.1.3 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
    100.00% packet loss

[SW1]
[SW1]
[SW1]ping 10.1.1.4
  PING 10.1.1.4: 56 data bytes, press CTRL_C to break
    Request time out
    Request time out
    Request time out
    Request time out
    Request time out

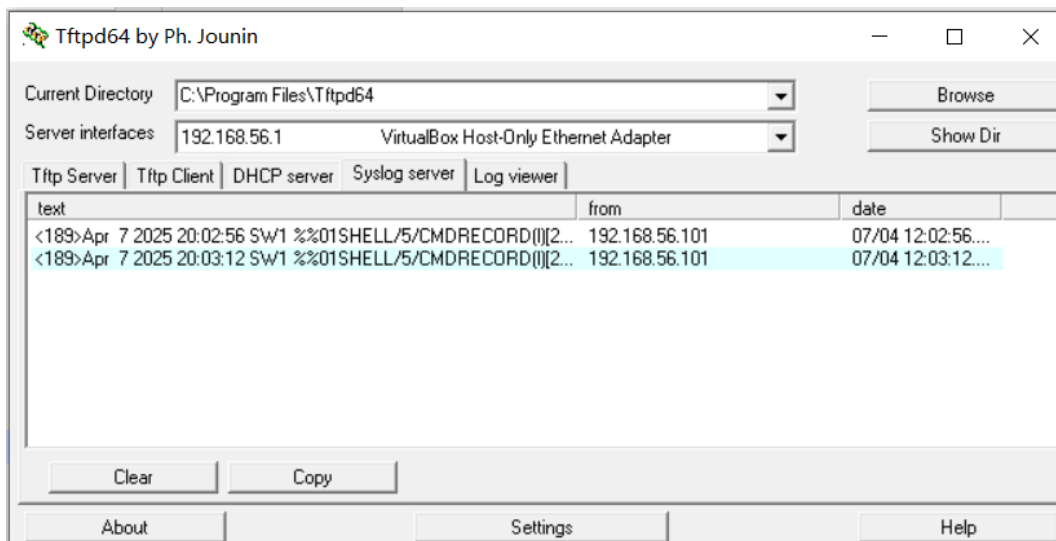
  --- 10.1.1.4 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
```



100.00% packet loss

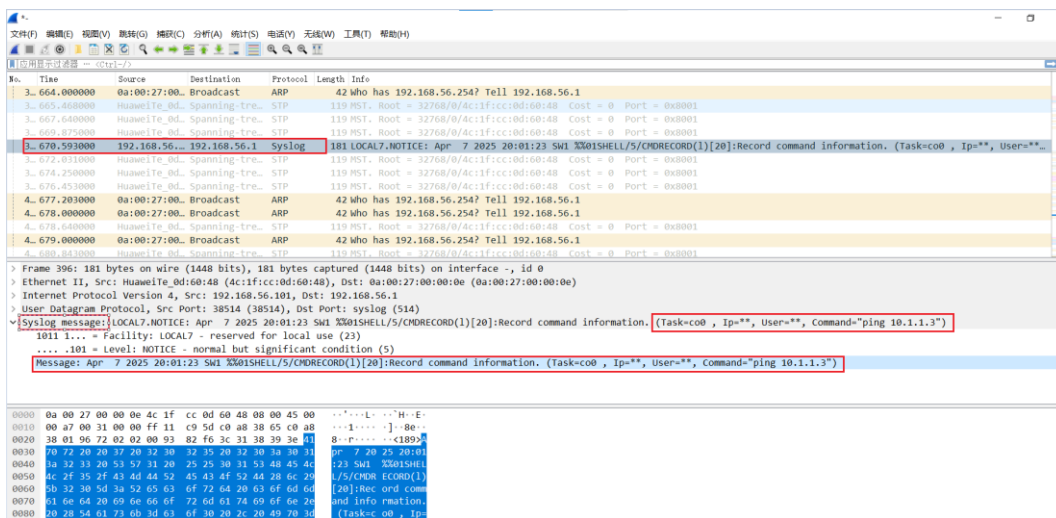
- (3) Check syslog information on Tftpd64 software, as shown in Figure 10.

Figure 10 Checking syslog information



- (4) Capture packets, as shown in Figure 11.

Figure 11 Capturing and checking syslog packets



4 Verification

Omitted.



5 Quiz

1. What is the default SNMP version in SNMP configuration?
By default, all SNMP versions (SNMPv1, SNMPv2c, and SNMPv3) are supported on Huawei AR G3 series routers.
2. What is the destination port number when the agent process sends trap messages to the NMS?
The agent process uses UDP to send trap messages to the NMS, and the destination port number is 162.
3. What is the main difference between SNMPv3 and SNMPv1/SNMPv2c, given their similar implementation principles?
SNMPv3 adds authentication and encryption mechanisms.